

NHS Satisfaction Analysis

British Social Attitudes Survey 2021 | R | Quantitative Research

1. Research question

Does NHS satisfaction differ across income, marital status, children, sex, education and age in the British Social Attitudes Survey 2021?

2. Data and analytical approach

The project uses the British Social Attitudes Survey 2021. NHS satisfaction is treated as a continuous outcome for parametric analysis, with a 0.05 significance level. The workflow uses labelled-variable recoding, complete-case analysis, descriptive summaries, an independent-samples t-test, a Chi-square test and simple linear regression.

3. Sample

The complete-case analytical sample reported in the project is $N = 3,307$. The mean NHS satisfaction score is 2.60. Non-substantive responses were excluded and age bands were represented using midpoints for the regression analysis.

4. Income comparison

Income band	Mean NHS satisfaction
Below £1,410	2.49
£1,411–£2,560	2.70
£2,561–£4,350	2.66
£4,351+	2.56

The observed means are close together, suggesting only modest differences in satisfaction across income groups.

5. Inferential results

Sex: the independent-samples t-test gave $t = -0.78$ and $p = 0.437$, so there is no statistically significant evidence of a difference in mean NHS satisfaction by sex at the 5% level.

Education: the Chi-square test gave $\chi^2 = 10.84$, $df = 6$, $p = 0.093$. This does not reach the 5% significance threshold.

Age: the simple linear regression estimated a slope of 0.00117 with $p = 0.717$ and approximately zero R^2 . There is therefore no evidence of a linear relationship between age and NHS satisfaction in this analysis.

6. Marital status and children

Descriptive means were 2.63 for married/cohabiting respondents, 2.52 for divorced/separated respondents, 2.88 for widowed respondents ($n = 16$, so interpret cautiously), and 2.55 for single respondents. Respondents with children averaged 2.56 versus 2.63 for those without children. These are descriptive comparisons rather than causal effects.

7. Limitations and conclusion

The analysis is cross-sectional and observational, so associations should not be interpreted causally. NHS satisfaction is measured on a discrete response scale even though it is analysed as continuous, and complete-case analysis can introduce selection bias. Overall, the examined socio-demographic variables provide weak evidence of systematic differences in NHS satisfaction in this sample.